Set 1. *Multi-step equations (Book 2 depth). Find x.*

1. $\frac{1}{2}(x+4) = x - 1$

2. $3(x-2) = 2(x+1)$

3. $\frac{2}{3}x + 1 = \frac{1}{3}x + 5$

4. $5 - 2(x-3) = x + 1$

5. $\frac{3}{4}(x-4) = 2x - 7$

6. $4(x+1) - 3 = 2(x+5)$

Set 2. *The "what's actually asked" trap – solve the equation, then answer the question actually asked, not just x.*

1. If $3x + 5 = 20$, what is $6x + 10$?

2. If $2(x-3) = 10$, what is $x - 3$?

3. If $4x - 7 = 9$, what is $2x$?

4. If $\frac{x+2}{3} = 5$, what is $x + 2$?

5. If $5x + 1 = 26$, what is $10x + 2$?

6. If $3(2x+1) = 21$, what is $2x + 1$?

Set 3. *Inequalities – watch for the sign flip. Find x.*

1. $2x + 3 < 11$

2. $-4x + 1 \geq 13$

3. $5 - x \leq 8$

4. $\frac{1}{2}x - 3 > 1$

5. $-3(x - 2) < 9$

6. $7 - 3x \geq -5$

Set 4. *Compound inequalities. Find x.*

1. $-2 < x + 3 \leq 7$

2. $3x - 1 < -4$ or $3x - 1 > 8$

3. $-5 \leq 2x + 1 < 9$

4. $x - 4 < -6$ or $x - 4 > 2$

5. $-1 < \frac{x + 1}{2} \leq 3$

6. $2x + 3 < 1$ or $2x + 3 > 9$

Set 5. *Translate the words into an inequality, then solve.*

1. A number increased by 5 is at least 12. (Find the number.)

2. Three times a number is no more than 21.

3. A number decreased by 4 is less than 10.

4. Twice a number, decreased by 3, is at least 7.

5. A number is at least -1 and at most 5. (Just write the compound inequality – no need to solve.)

6. Five less than twice a number is no more than 15. What is the largest possible integer value of the number?

Set 6. *Spot the error – name the mistake, give the correct answer.*

1. In solving $-2x > 8$, a student wrote:
   "Divide both sides by -2: $x > -4$."
   Error + fix: . . . . . .

2. In solving $x - 5 > 2$, a student wrote:
   "There's a negative number, so flip the sign: $x < 7$."
   Error + fix: . . . . . .

3. For "if $4x - 1 = 15$, what is $4x$?" a student wrote:
   "$x = 4$, so the answer is 4."
   Error + fix: . . . . . .

4. In solving $-2(x - 3) < 10$, a student wrote:
   "Distribute: $-2x - 6 < 10$, so $x > -8$."
   Error + fix: . . . . . .

5. In solving $-4 < 2x < 8$, a student wrote:
   "Divide the middle and right side by 2: $-4 < x < 4$."
   Error + fix: . . . . . .

Set 7. *Bonus / challenge – the hardest items in this set.*

1 If $\frac{2}{3}(x-3) = x-5$, what is the value of x? What is the value of $3x$?

2 Solve $-5 < 3x + 1 \leq 10$. What is the resulting compound inequality, and what is the smallest integer value of x that satisfies it?

3 Seven minus twice a number is at least -3. What is the largest possible value of the number?

4 If $4(x-1) - 2x = 3(x+2) - 10$, what is the value of x? What is the value of $x + 2$?

5 What is the solution set of $-6 \leq -2(x-1) < 4$?

Set 8. *Official SAT-style application (next 5 pages) – 45 original, multiple-choice problems written to match the method of real, officially-released SAT Suite Question Bank items for this lesson's two skill categories: Linear equations in one variable, and Linear inequalities in one or two variables. 15 Easy, 15 Medium, 15 Hard – matching real-test density. No real College Board question is reproduced; see the sourcing note after Set 8's last page.*

Set 8 – Easy. *Choose the correct answer.*

1 If $5x = 12$, what is the value of $20x$?

(A) 3 (B) 20

(C) 48 (D) 60

2 If $4 + x = 15$, what is the value of $12 + 3x$?

(A) 33 (B) 41

(C) 45 (D) 48

3 How many solutions does the equation $3(x+2) = 3x+6$ have?

(A) None (B) Exactly one

(C) Exactly three (D) Infinitely many

4 How many solutions does the equation $x + 8 = x + 3$ have?

(A) None (B) Exactly one

(C) Exactly three (D) Infinitely many

5 What is the solution to the equation $5x - 4 = 11$?

(A) 1 (B) 3

(C) 5 (D) 15

6 A bakery sells muffins for \$3 each and loaves of bread for \$5 each. On Monday, the bakery sold m muffins and 12 loaves of bread, and the total sales were \$111. Which equation represents this situation?

(A) $3m + 5(12) = 111$

(B) $3(12) + 5m = 111$

(C) $3m + 5m = 111$

(D) $m + 12 = 111$

Set 8 – Easy (cont.). *Choose the correct answer.*

7 What is the solution to the equation $3(x - 2) = 2x + 5$?

(A) 1 (B) 7

(C) 11 (D) 17

8 What is the solution to the equation $4x - 5 = 19$?

(A) 3 (B) 6

(C) 24 (D) 96

9 A charity needs to collect at least 150 canned goods. If the charity has already collected 118 cans, what is the minimum number of additional cans needed?

(A) 268

(B) 118

(C) 32

(D) 0

10 What is the solution set of the inequality $3x - 7 \leq 14$?

(A) $x \leq 7$ (B) $x \geq 7$

(C) $x \leq 21$ (D) $x \leq -7$

11 A food truck sells only tacos and burritos and can prepare at most 80 items per day. If t is the number of tacos and b is the number of burritos prepared in a day, which inequality represents this situation?

(A) $t + b \leq 80$

(B) $t + b \geq 80$

(C) $t - b \leq 80$

(D) $t - b \geq 80$

12 A rental company charges a \$25 delivery fee plus \$2 per chair. Diyora wants to spend no more than \$95 total. What is the maximum number of chairs she can rent?

(A) 30

(B) 33

(C) 35

(D) 60

Set 8 – Easy (cont.). *Choose the correct answer.*

13 What is the solution to the equation $1.5x + 3 = 12$?

(A) 4 (B) 6

(C) 9 (D) 13.5

14 What is the solution to the equation $\frac{x}{4} + 2 = 9$?

(A) 7 (B) 11

(C) 28 (D) 36

15 If $2x + 1 \geq 15$, what is the smallest possible integer value of x?

(A) 6 (B) 7

(C) 14 (D) 15

Set 8 – Medium. *Choose the correct answer.*

1 What is the solution to the equation $5(x-2) = 3(x+4)$?

(A) 1 (B) 4

(C) 11 (D) 22

2 What is the solution to the equation $\frac{1}{2}(x-3)-\frac{1}{4}(x-3) = 5$?

(A) 3 (B) 17

(C) 20 (D) 23

3 In the equation $a(x-4) - b = 2x - 9$, a and b are constants. If the equation has infinitely many solutions, what are the values of a and b?

(A) $a = 2,\ b = 1$

(B) $a = 2,\ b = -1$

(C) $a = -2,\ b = 1$

(D) $a = -2,\ b = -1$

4 A rectangular banner has width w feet; its length is 5 feet more than twice its width. Which expresses the perimeter, in feet, in terms of w?

(A) $3w + 5$ (B) $6w + 5$

(C) $6w + 10$ (D) $2w^2 + 5w$

5 If $47 = 15 + 4(x - 2)$, what is the value of $4(x - 2)$?

(A) 8 (B) 15

(C) 32 (D) 47

Set 8 – Medium (cont.). *Choose the correct answer.*

6 A delivery van can carry a load if the combined weight of the cargo and the packaging is no more than 3,800 pounds. The packaging weighs 200 pounds and each package weighs 85 pounds. What is the maximum number of packages the van can carry?

(A) 42

(B) 43

(C) 45

(D) 47

7 What is the solution set of $-5 < 2x + 1 \leq 9$?

(A) $-3 < x \leq 4$ (B) $-6 < x \leq 8$

(C) $-2 < x \leq 5$ (D) $-3 \leq x < 4$

8 For which of the following tables are all the values of x and their corresponding values of y solutions to $y < 4x+3$?

(A) $(2, 11),\ (4, 19),\ (6, 27)$

(B) $(2, 10),\ (4, 20),\ (6, 24)$

(C) $(2, 12),\ (4, 16),\ (6, 20)$

(D) $(2, 8),\ (4, 16),\ (6, 24)$

9 If $\frac{x+4}{3} = \frac{x+4}{7}$, what is the value of $x + 4$?

(A) -4 (B) 0

(C) 4 (D) 7

Set 8 – Medium (cont.). *Choose the correct answer.*

10 If $11 < 2n + 3 \leq 21$, which of the following gives all possible values of n?

(A) $4 < n \leq 9$

(B) $4 \leq n < 9$

(C) $8 < n \leq 18$

(D) $7 < n \leq 18$

11 What is the solution to the equation $2.5(x+4) = 3x-1$?

(A) 11 (B) 18

(C) 22 (D) 44

12 A print shop prints small, medium, and large posters. The number of medium posters is 3 times the number x of small posters, and the number of large posters is 15. The shop printed 95 posters total that day. Which equation represents this situation?

(A) $x(3)x + 15 = 95$

(B) $x + 3x + 15x = 95$

(C) $3x + 15 = 95$

(D) $4x + 15 = 95$

13 A line in the xy-plane has a y-intercept of -2 and a slope of 5. The solution region for a certain inequality is all points above this line. Which inequality represents this region?

(A) $y > 5x - 2$

(B) $y < 5x - 2$

(C) $y > -5x - 2$

(D) $y > 5x + 2$

14 If $6x - 2(x - 3) = 3(x + 1) + 1$, what is the value of $2x$?

(A) -4 (B) -2

(C) 2 (D) 4

15 For what value of k does $5x + 9 = kx + 9 + 2x$ have infinitely many solutions?

(A) 2 (B) 3

(C) 5 (D) 7

Set 8 – Hard. *Choose the correct answer.*

1 A bakery makes small, medium, and large bread loaves. The number of large loaves made is 6 times the number n of medium loaves, and the number of small loaves made is 18. The bakery makes 150 loaves total that day. Which equation represents this situation?

(A) $18(6n) + n + 18 = 150$

(B) $6n + 6(18) = 150$

(C) $n + 6n + 18n = 150$

(D) $7n + 18 = 150$

2 If $\frac{x+3}{6} = \frac{x+3}{10}$, the value of $x + 3$ is between which of the following pairs of values?

(A) -8 and -6

(B) -3 and 3

(C) 5 and 7

(D) 9 and 12

3 In the given equation, k and n are constants and $n > 2$:

$$3(kx - n) = -\frac{5}{6}x - \frac{9}{4}$$

If the equation has no solution, what is the value of k?

(A) $-\frac{5}{18}$

(B) $\frac{5}{18}$

(C) $-\frac{3}{4}$

(D) $-\frac{9}{4}$

Set 8 – Hard (cont.). *Choose the correct answer.*

4 A shipping company requires that for boxes shaped like rectangular prisms, the sum of the base's perimeter and the box's height cannot exceed 150 inches. A box has height 50 inches and length 3 times its width. Which inequality shows the allowable width x, in inches?

(A) $0 < x \leq 10$

(B) $0 < x \leq 12.5$

(C) $0 < x \leq 25$

(D) $0 < x \leq 100$

5 The solution region for $ry > 84$, where r is a constant, consists of all points *below* the horizontal line $y = -6$ in the xy-plane (not including the line). What is the value of r?

(A) 14

(B) -14

(C) 6

(D) -6

6 For which of the following tables are all the values of x and their corresponding values of y solutions to $y \leq 9x - 25$?

(A) $(3, 3),\ (6, 25),\ (9, 50)$

(B) $(3, -5),\ (6, 30),\ (9, 50)$

(C) $(3, -5),\ (6, 20),\ (9, 57)$

(D) $(3, -5),\ (6, 20),\ (9, 50)$

7 Akmal is 3 years more than twice his sister Malika's current age. In 5 years, the sum of their ages will be 37. How old is Akmal right now?

(A) 8

(B) 16

(C) 19

(D) 21

Set 8 – Hard (cont.). *Choose the correct answer.*

8 What value of x satisfies $\frac{x}{3} - \frac{x}{5} = 8$?

(A) 8 (B) 15

(C) 24 (D) 60

9 If $-11 \leq 3(x - 2) < 16$, what is the greatest possible integer value of x?

(A) 6 (B) 7

(C) 8 (D) 22

10 In the equation $4x + c = d(x - 3) + 7$, c and d are constants. If the equation has infinitely many solutions, what is the value of $c + d$?

(A) -9

(B) -1

(C) 3

(D) 11

11 The sum of three consecutive even integers is 54 more than twice the smallest of the three integers. What is the largest of the three integers?

(A) 48

(B) 50

(C) 52

(D) 150

Set 8 – Hard (cont.). *Choose the correct answer.*

12 A rectangular garden's length is 7 feet less than 3 times its width. The perimeter of the garden is 106 feet. What is the length, in feet, of the garden?

(A) 15

(B) 23

(C) 38

(D) 45

13 What value of x satisfies $\frac{3}{4}(x + 8) = \frac{1}{2}x + 10$?

(A) 4 (B) 8

(C) 16 (D) 24

14 A concert venue sells tickets for \$45 each and charges a one-time \$120 cleaning fee for a private host. The host has a budget of \$3,200 for tickets and the fee. What is the maximum number of tickets the host can purchase?

(A) 66

(B) 68

(C) 69

(D) 71

15 For what value of the constant a is $a(x + 4) = 3x + 12$ true for every value of x?

(A) 1

(B) 3

(C) 4

(D) 12

Sourcing note: every item in Set 8 is original – written by studying the method and structure of real, officially-released SAT Suite Question Bank items in this lesson's skill categories, then building a new problem (new numbers, names, and context) requiring the same solving method. No real College Board question text is reproduced anywhere in this book.

Answer Key – Sets 1–7

Set	#	Answer	#	Answer
Set 1	1	$x = 6$	2	$x = 8$
	3	$x = 12$	4	$x = \frac{10}{3}$
	5	$x = \frac{16}{5}$	6	$x = \frac{9}{2}$
Set 2	1	40	2	5
	3	8	4	15
	5	52	6	7
Set 3	1	$x < 4$	2	$x \leq -3$
	3	$x \geq -3$	4	$x > 8$
	5	$x > -1$	6	$x \leq 4$
Set 4	1	$-5 < x \leq 4$	2	$x < -1$ or $x > 3$
	3	$-3 \leq x < 4$	4	$x < -2$ or $x > 6$
	5	$-3 < x \leq 5$	6	$x < -1$ or $x > 3$
Set 5	1. $x + 5 \geq 12 \Rightarrow x \geq 7$			
	2. $3x \leq 21 \Rightarrow x \leq 7$			
	3. $x - 4 < 10 \Rightarrow x < 14$			
	4. $2x - 3 \geq 7 \Rightarrow x \geq 5$			
	5. $-1 \leq x \leq 5$			
	6. $2x - 5 \leq 15 \Rightarrow x \leq 10$, largest integer $= 10$			
Set 6	1. Error – dividing by a negative should flip the sign, but the direction shown is backwards. Correct: $x < -4$.			
	2. Error – no multiplication/division by a negative happened (only subtraction); sign should not flip. Correct: $x > 7$.			
	3. Error – reported x instead of $4x$. Correct: $x = 4$, so $4x = 16$.			
	4. Error – distributed $-2 \times (-3)$ as -6 instead of $+6$. Correct: $-2x + 6 < 10 \Rightarrow x > -2$.			
	5. Error – divided the middle and right side by 2 but left -4 un-divided. Correct: $-2 < x < 4$.			
Set 7	1. $x = 9$, $3x = 27$			
	2. $-2 < x \leq 3$, smallest integer $= -1$			
	3. $x \leq 5$, largest value $= 5$			
	4. $x = 0$, $x + 2 = 2$			
	5. $-1 < x \leq 4$			

Answer Key – Set 8

Easy

1	2	3	4	5	6	7	8	9	10
C	C	D	A	B	A	C	B	C	A

11	12	13	14	15
A	C	B	C	B

Medium

1	2	3	4	5	6	7	8	9	10
C	D	A	C	C	A	A	D	B	A

11	12	13	14	15
C	D	A	A	B

Hard

1	2	3	4	5	6	7	8	9	10
D	B	A	B	B	D	C	D	B	B

11	12	13	14	15
C	C	C	B	B

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